Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad, Natore

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lecture Plan

Program:	**B. Sc. in Computer Science and Engineering (CSE)**
Course Code & Title:	EEE-1161 (Electrical Circuit Analysis)
Semester:	Summer-2024
Level:	1st Year 1st Semester
Credit Hour:	3.00 Hrs./Week
Name & Designation of Teacher:	Ashik Alam Assistant Professor, Dept. of EEE, BAUET
Class Hours:	Sunday: 11:40 AM - 12:35 PM Monday: 10:15 AM - 11:10 AM Wednesday: 10:15 AM - 11:10 AM
Consultation Hours:	Thursday: 9:15 AM - 11:10 AM
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Rationale:	To learn and familiarize the basics of electrical circuit components as well as the analysis of DC and AC circuits
Pre-requisite (if any):	None

Course Objective:
This course intends to give idea about the basic electrical quantities with regards to both DC and AC, techniques to solve electrical circuits, applications of network theorem and magnetic circuits.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

CO1	Understand the concept of circuit, classification of elements and types of energy sources.
CO2	Apply different laws of circuit theorems to solve various engineering problems.

Course Content:

Basic Knowledge of Electrical Circuit, Measuring the Quantities: Current, Voltage, Resistance.

Standards and Units of Electrical Technology, Network & Circuit Solution: Series, parallel, node and mesh analysis.

Instruments of Measurement: Series and parallel circuits; Network theorem and network analysis; Capacitors, Inductors, Basic of magnetic circuits.

AC Circuit Analysis: Instantaneous current, voltage and power for RLC circuits, Effective current and voltage, average power, Phasor representation of sinusoidal quantities, Single phase circuit, Introduction of three-phase circuits; Power factor and power equation

Text and Ref Books:

1. Fundamentals of Electric Circuits - Charles K. Alexander and Mathew N. O. Sadiku
2. Introductory Circuit Analysis - Robert L. Boylestad
3. Alternating-current Circuits - G. F. Corcoran and R. M. Kerchner

***Additional reading material:** Power Point slides & PDF provided by the teacher

Minimum Attendance:

80% class attendance is mandatory for a student in order to appear at the final examination.

Student's Responsibilities:

1. Students must come to the class prepared for the course material covered in the previous class
2. They must submit their assignments on time
3. No late or partial assignments will be acceptable
4. There will be no make-up Class Test

Detailed Mapping (According to BAETE Accreditation Manual- II)

Course Code & Title: EEE 1101 (Electrical Circuits I)

Mapping CO-PO, Bloom's Level, Knowledge Profile, Complex Engineering Problem (WP) and Complex Engineering Activities (EA)

COs	**CO Statements** (Upon completion the course students will be able to)	**PO**	**WK**	**WP**	**EA**	**Bloom's Level**	**Delivery methods and activities**	**Assessment Tools**
CO1	Understand the concept of circuit, classification of elements and types of energy sources.	PO1	WK2 WK3	-	-	C2	Lectures, Text Book, PPT,PDF Handout Tutorial	CP / Presentation, Assignment, CT, SF
CO2	Apply different laws of circuit theorems to solve various engineering problems.	PO2	WK2 WK3	-	-	C3	Lectures, Text Book, PPT,PDF Handout Tutorial	CP / Presentation, Assignment, CT, SF

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive Domain (C1-C6), Psychomotor Domain (P1-P7), and Affective Domain (A1-A5)

	Program Outcome (PO)											
COs	1	2	3	4	5	6	7	8	9	10	11	12
CO1	M											
CO2		M										

Legends: High (H)=3, Medium(M)=2 and Low(L)=1

Teaching and Learning Strategies (TLS)

TLS1 Delivery lectures using multimedia of different topics

TLS2 Active and fruitful discussion in class regarding efficient solving of the logical, mathematical and complex engineering problems

TLS3 Group discussion and presentation regarding diverse problems and corresponding lectures

TLS4 Evaluation of class performance to reach each student in a class for every topic.

Detail Lecture Planning

SL No.	Lecture Number	Content	Remarks
1.	Lecture 1	Introduction, Familiarization, Marks distribution, Electrical and Electronic Basic Knowledge	
2.	Lecture 2	**Measuring the Quantities:** Current, Voltage, Resistance.	
3.	Lecture 3	power, energy	
4.	Lecture 4	Basic Laws: Ohm's law, Kirchhoff's current and voltage laws	
5.	Lecture 5	Voltage divider and current divider rules	
6.	Lecture 6	Series, parallel and series-parallel circuits and their Equivalents,	
7.	Lecture 7	Delta-Wye or vice versa conversion technique and Delta-Wye equivalent circuits	
8.	Lecture 8	independent and dependent sources, mesh analysis including super mesh	
9.	Lecture 9	**Revision (Discussion & Problem-solving)**	
10.	Lecture 10	Nodal analysis including super node	
11.	Lecture 11	Nodal analysis (dependent sources)	
12.	Lecture 12	Network Theorem: Thevenin's theorem	
13.	Lecture 13	Network Theorem: Thevenin's theorem (dependent sources)	
14.	Lecture 14	Norton's theorem	
15.	Lecture 15	Norton's theorem including dependent source	
16.	Lecture 16	superposition theorem with applications in circuits having independent and dependent sources	
17.	Lecture 17	Maximum power transfer theorem for DC load	
18.	Lecture 18	Source Transformation Theorem.	
19.	Lecture 19	Sinusoidal Functions: Instantaneous current, voltage, power, effective current and voltage,	
20.	Lecture 20	Average power, phasor and complex quantities.	
21.	Lecture 21	Impedance, admittance, reactance, susceptance of RL, RC and RLC branches.	
22.	Lecture 22	non-sinusoidal waveforms, calculation of RMS and average value for non-sinusoidal waveforms	
23.	Lecture 23	Analysis of Single-Phase AC Circuits: Vector diagram representation of AC circuits	
24.	Lecture 24	Series and parallel RL, RC and RLC circuits	
25.	Lecture 25	**Revision**	
26.	Lecture 26	Techniques of general ac circuit analysis (containing both independent and dependent sources)	

27.	Lecture 27	nodal analysis for AC circuits	
28.	Lecture 28	mesh analysis for AC circuits	
29.	Lecture 29	superposition theorem and Source Transformation Theorem for AC circuits	
30.	Lecture 30	Maximum power transfer theorem for AC load	
31.	Lecture 31	Basic of magnetic circuits	
32.	Lecture 32	Introduction to three phase circuits; (Δ and Y circuits).	
33.	Lecture 33	Introduction to three phase circuits; Power factor and power equation (Δ and Y circuits).	
34.	Lecture 34	Power factor and power equation	
35.	Lecture 35	three phase circuits power calculation	
36.	Lecture 36	Energy Storage Elements: Properties of Inductances and capacitances,	
37.	Lecture 37	Series-parallel combinations of inductances and capacitances	
38.	Lecture 38	Responses of RL and RC circuits	
39.	Lecture 39	**Revision**	
40.	Lecture 40	**Whole syllabus revision, Question Pattern and Answering Policy for good result**	
41.	Lecture 41	**Reserve day 1**	**for any vacation issue**
42.	Lecture 42	**Reserve day 2**	**for any vacation issue**

Prepared by: **Approved by: (Head of the Dept.)**

Ashik Alam
Assistant Professor,
Dept. of EEE, BAUET